EM-Based Transfer Learning for Gaussian Causal Models Under Covariate and Target Shift

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***Abstract*—We present a unified EM-based framework for imputing a designated target variable T in Gaussian causal models under domain shifts. Given fully observed source-domain data and only partially observed target-domain data, our method flexibly handles both covariate and target shifts by exploiting a known Directed Acyclic Graph (DAG) structure to transfer information from observed data to the missing target. Our framework introduces: (i) a population EM operator that characterizes the exact covariance update in the infinite-data regime, (ii) a first-order gradient–EM step that reduces per-iteration cost from $O(p^3)$ to $O(p^2)$ while preserving likelihood ascent, and (iii) a sample-EM routine that adapts to domain shifts by freezing source-invariant parameters and re-estimating only the shifted conditionals. In experiments on a synthetic seven-node Structural Equation Model (SEM), the 64-node MAGIC-IRRI network, and the real single-cell signaling dataset of Sachs et al., our 1st-order EM achieves near-oracle accuracy under severe shifts in the small model, markedly outperforms the classical Kiiveri EM and a fit-on-source Bayesian network baseline in the large model, and demonstrates strong gains on key signaling proteins in real biological data. Because it performs just one EM update and scales linearly with the number of edges, our approach is both computationally efficient and practical for high-dimensional Gaussian causal models.**

***Index Terms*—Data Shift, EM algorithm, Causality, DAG, Gaussian SEM, Missing Data.**

I. Introduction

Domain Adaptation. Domain adaptation studies how to transfer predictive models learned in a *source* domain to a *target* domain whose data distribution differs. Two canonical shifts have been discussed in the literature:

1) **Covariate shift** occurs when the marginals of the *context* variables differ between source and target, while the conditional $P(Y \mid X)$ remains invariant [1]–[3].
2) **Target shift** (or *label shift*) arises when the marginal of the *target* variable changes across domains, but $P(X \mid Y)$ is unchanged [4]–[6].

For an overview of additional domain adaptation scenarios and theoretical results, we refer the reader to [7]. In this work, we focus primarily on covariate, target, and generalized target shifts where the marginal $P(Y)$ changes, and the conditional $P(X|Y)$ changes with constraints.

Causal Inference for Domain Adaptation. Causal methods can exploit the underlying cause–effect structure in the data to guard against distributional shifts [8]–[14]. Key approaches include:

- **Transportability** formalizes differences and commonalities between populations via *selection diagrams*, using do-calculus [15] to decide when interventional or observational effects can be carried over [16]–[19].
- **Invariant causal prediction** (ICP) seeks subsets of predictors whose regression residuals exhibit identical distributions across environments [20]–[22]. Identifiability in the nonlinear or partially observed case remains an open problem [23].
- **Graph surgery** removes unstable mechanisms from the factorization to enforce cross-domain invariance [24], [25].
- **Graph pruning** frames adaptation as selecting the optimal subset of predictors yielding invariant conditionals [26]–[29].

However, even when a subset A can be found that guarantees zero transfer bias (for instance via graph pruning), the resulting incomplete-information bias may still yield large prediction errors. Alternatively, causal approaches such as graph surgery require prior estimation of causal effects or counterfactual reasoning—adding considerable complexity—and most of these methods suffer from limited scalability. More recent advances in *invariant risk minimization* [30] and deep generative approaches for causal representation learning [31], [32] promise improved robustness to unseen shifts. In this paper, we take a different tack: we show that, under a linear–Gaussian SEM, an EM-based estimator provably converges geometrically to a good fixed point that recovers the causal model of the target domain even when all target predictions are missing.

A Motivating Example. We work with the linear Gaussian SEM whose causal structure is depicted in Fig. 2. The seven nodes consist of two *context* or background variables C_1, C_2, two intermediate features Z, X, the *target* variable T, and two downstream outcomes P, Y. Concretely,

$$
\begin{aligned}
&Z = 2\,C_1 + 3\,C_2 + \varepsilon_Z, \quad X = 3\,C_1 + \varepsilon_X,\\
&T = \beta_{C_1\to T}\,C_1 + \beta_{X\to T}\,X + \beta_{Z\to T}\,Z + \varepsilon_T, \quad P = T + \varepsilon_P,\\
&Y = 2\,T + \varepsilon_Y,
\end{aligned}
$$

with all noise terms $\varepsilon_\bullet \sim \mathcal{N}(0,1)$ i.i.d. In the *source* domain we draw each context variable $C_i \sim \mathcal{N}(0,1)$; in the *target* domain we introduce two forms of shift:

- **Covariate shift** by shifting the marginal of C_2 (e.g. $C_2 \sim \mathcal{N}(\mu_{\mathrm{tgt}}, \sigma^2_{tgt})$),

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- **Generalized target shift** by shifting both the marginal of C_1 and the prior $P(T)$ via an added constant or inflated noise variance in the ε_T term.

Although T is completely unobserved in the target domain, the DAG structure remains invariant across domains. We will compare three methods for predicting (or imputing) T under these shifts: (a) a "fit-on-source" Bayesian network baseline, (b) the classic Kiiveri EM algorithm [33] treating T as latent, and (c) our proposed first-order EM algorithm (see Sect. IV). Subsequent results appear in Table 1 and Fig. 1.

TABLE 1. Average error metrics under covariate and target shift for the motivating example.

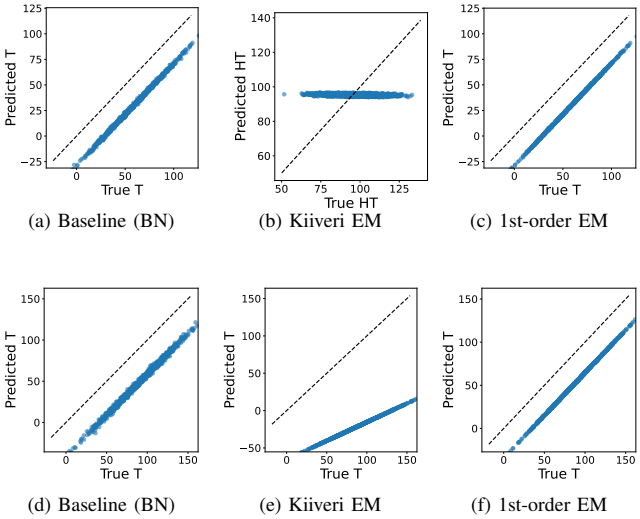
Shift scenario	Method	MAE	RMSE	R^2
Covariate shift	Baseline (Fit-on-Source)	0.7962	1.0137	0.9981
	Kiiveri EM	45.4529	45.4554	–2.8735
	1st-order EM	**0.3331**	**0.4273**	**0.9997**
Target shift	Baseline (Fit-on-Source)	6.1528	6.4643	0.9471
	Kiiveri EM	71.8909	73.2513	–5.7872
	1st-order EM	**1.0386**	**1.1228**	**0.9984**

Discussion. Under covariate shift (shifting C_2 only) and generalized target shift (shifting both C_1 and $P(T)$), the naive baseline—training a Bayesian network on the source and directly predicting—suffers only mildly under covariate shift but degrades substantially under target shift. The classic Kiiveri EM algorithm, while known to be *insensitive to initialization* and reliably *convergent* even from poor starting points [33, p. 547], here converges to boundary or saddle-point solutions with very large errors in both shift scenarios. In contrast, our first-order EM, by initializing at the source-domain covariance and exploiting the known DAG structure, converges to the correct fixed point (despite T being entirely missing in the target) and achieves near-ideal error metrics under both covariate and target shifts. Moreover, while EM is known to enjoy statistical guarantees of geometric convergence up to a statistical precision neighborhood of the optimum [34], these results hold for specific canonical settings—mixtures of two Gaussians, mixtures of two regressions, and linear regression with covariates missing completely at random—not for Gaussian causal models as considered here. In this motivating example, our first-order EM consistently exhibits monotone ascent of the observed likelihood and geometric decay of successive updates[1], providing strong empirical evidence of convergence to a stable fixed point with near-oracle predictive accuracy. This simple method thus offers a scalable and practically robust solution to domain adaptation.

Summary of Contributions Our work provides:

- **Population-EM operator for Gaussian DAGs** (Section IV-A): we derive a closed-form fixed-point mapping F which, in the infinite-data regime, updates the full covariance by the exact conditional second moment of the complete data.

[1]Empirical convergence diagnostics, including likelihood trajectories and successive-gap plots, are provided in the supplementary material at our GitHub repository.



(a) Baseline (BN) (b) Kiiveri EM (c) 1st-order EM

(d) Baseline (BN) (e) Kiiveri EM (f) 1st-order EM

Fig. 1. True vs. predicted (imputed) T under covariate shift (top row) and generalized target shift (bottom row). The 1st-order EM uniformly achieves near-perfect recovery of T across both shift types, while the baseline and Kiiveri EM fail catastrophically under severe domain shift.

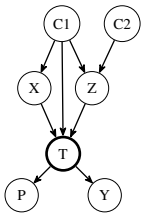


Fig. 2. Causal DAG underlying the shared data-generating process across source and target domains for the motivating example. $\mathbf{C}_1$, $\mathbf{C}_2$: context; **Z**, **X**: intermediates; **T**: target (unobserved in target domain); **P**, **Y**: downstream. The causal structure is invariant across domains.

- **First-order (partial) M–step via gradient-EM** (Section IV-B): we replace the $O(p^3)$ generalized least squares (GLS)-based M–step by a single gradient-ascent update on Σ, reducing per-iteration cost to $O(p^2)$ while still guaranteeing ascent in the expected complete-data log-likelihood.
- **Domain-Adaptive EM** (Section IV-C): we develop an EM routine that "freezes" source-invariant parameters, computes an E–step on target-observed data, and re-estimates only the shifted conditional(s), yielding an $O(p^2)$ per-iteration algorithm tailored to domain adaptation.
- **Geometric convergence guarantees** (Sections IV-D–IV-F): building on Balakrishnan–Wainwright–Yu, we prove local contraction of the population operator (Sec. IV-D), decompose the observed-data Hessian into complete- vs. missing-data components (Sec. IV-E), derive explicit spectral-gap conditions under covariate/target shift (Sec. IV-E), and extend to high-probability sample-level error bounds (Sec. IV-F).

II. Related Work

a) The Classical EM Algorithm and Early Variants: Dempster, Laird, and Rubin established the EM algorithm as a general-purpose method for maximum-likelihood estimation with incomplete data, proving that each iteration increases the

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observed-data likelihood and that EM converges to a stationary point under mild conditions [35]. Louis [36] provided the missing-information decomposition of the observed-data Hessian into a strictly negative-definite complete-data term minus a positive semidefinite missing-data correction, elucidating local curvature conditions for EM's convergence. Wu [37] proved that EM converges to a local optimum and analyzed its asymptotic linear convergence rate. Subsequent surveys and monographs (e.g., [38]) have collected these early convergence guarantees and described practical issues such as initialization and local maxima.

b) EM for Covariance Structure and Structural Equation Models: Early work in psychometrics and structural equation modeling recognized that Gaussian SEMs with latent or missing variables naturally fit into the EM framework. Jöreskog and Sörbom [39] and McArdle and McDonald [40] describe covariance-structure estimation via iterative algorithms, but Kiiveri [33] systematized the incomplete-data viewpoint by deriving closed-form expressions for $E[XX^T \mid X_{\text{obs}}]$ and for the corresponding score and Hessian. This allowed efficient EM (and Newton-type) updates for fitting recursive and factor-analysis models with missing entries or latent constructs.

c) EM Variants, First-Order EM, and Tutorial Overviews: A rich collection of EM variants has been proposed to improve computational efficiency or convergence speed. Meng and Rubin [41] introduced ECM, which decomposes the M-step into simpler conditional maximizations, and Liu and Rubin [42] developed ECME, which further accelerates convergence by maximizing either the observed or expected complete-data log-likelihood when convenient. Meng and Van Dyk's PX-EM [43] uses parameter expansion to improve curvature. Roche [44] provides a tutorial on EM and its many extensions, including GEM, ECM, ECME, Monte Carlo EM, and stochastic EM. More recently, Balakrishnan *et al.* [34] proposed a first-order EM (gradient EM) that replaces the exact M-step with a single gradient ascent, reducing per-iteration cost from $O(p^3)$ to $O(p^2)$ in Gaussian models.

d) Modern Statistical Guarantees- Population vs. Sample-Level Analyses: Balakrishnan, Wainwright, and Yu [34] developed a unified framework showing that the EM operator in the infinite-data limit is a local contraction under a spectral-gap condition on complete vs. missing-data Hessians, and they derived nonasymptotic, high-probability bounds guaranteeing that sample EM converges geometrically to within $O(\sqrt{d/n})$ of the true parameter. Their analyses cover both exact M-step EM and first-order EM variants. Wang, Xu, and Ravikumar [45] extended these ideas to high-dimensional Gaussian mixtures, showing that a truncated EM algorithm can recover a sparse mean vector at minimax-optimal rates when $d \gg n$.

e) Other Notable EM-Related Advances.: Extensions of EM to large-scale and streaming settings include online EM [46], which replaces full E-steps with stochastic approximations, and stochastic EM variants that process mini-batches or single samples [47]. Monte Carlo EM (MCEM) and Stochastic EM [47], [48] use Monte Carlo or MCMC to approximate intractable E-steps. Variational EM (VBEM) extends EM to approximate Bayesian inference by optimizing a lower bound on the marginal likelihood [49]. Collectively, these advances enable EM-style updates in scenarios ranging from massive data to complex latent-variable models.

III. Problem Statement

We first formalize the domain-adaptation task and then outline how to construct a population-EM operator whose sample-based counterpart admits geometric convergence guarantees.

Let $X = (X_1, \ldots, X_p)$ be p random variables with a known causal DAG $\mathcal{G}$. One coordinate $T = X_t$ is the *target variable*, fully observed in the source domain but entirely missing in the target domain. We write

$$\mathcal{D}_{\text{s}} \;=\; \big\{\, X_{\text{s}}^{(i)} \big\}_{i=1}^{N_{\text{s}}}, \quad X_{\text{s}}^{(i)} = \big(X_{1,\text{s}}^{(i)}, \ldots, X_{p,\text{s}}^{(i)}\big),$$

for the complete source-domain samples, and

$$\mathcal{D}_{\text{t}}^{\text{obs}} \;=\; \big\{\, X_{\text{t},-t}^{(j)} \big\}_{j=1}^{N_{\text{t}}},$$

$$X_{\text{t},-t}^{(j)} = \big(X_{1,\text{t}}^{(j)}, \ldots, X_{t-1,\text{t}}^{(j)}, X_{t+1,\text{t}}^{(j)}, \ldots, X_{p,\text{t}}^{(j)}\big),$$

for the observed target-domain samples (with T missing). Equivalently, $X_{\text{t},-t}^{(j)}$ collects all $p-1$ coordinates except X_t. We assume:

1) The source samples $X_{\text{s}}^{(i)}$ are fully observed across all p coordinates, including X_t.
2) In the target domain, $X_t^{(j)}$ is missing for all j; the remaining $p-1$ coordinates X_k (for $k \neq t$) are observed.
3) The DAG $\mathcal{G}$ (with adjacency matrix A) is known and invariant across both domains. Each node X_k follows a linear Gaussian structural equation:

$$X_k \;=\; \sum_{j \in \text{pa}(k)} \theta_{kj}\, X_j \;+\; \varepsilon_k, \quad \varepsilon_k \sim \mathcal{N}(0, \sigma_k^2).$$

   We collect the noise variances into $\Delta = (\sigma_1^2, \ldots, \sigma_p^2)^T$. In matrix form, one can write $A_{ii} = 1$, $A_{ij} = -\theta_{ij}$ if $j \in \text{pa}(i)$, and $A_{ij} = 0$ otherwise. The structural coefficients θ_{kj} are not set arbitrarily but are determined by the underlying data-generating process or inferred from covariance structure, and a detailed tutorial on parameter identification in linear structural equation models is provided by Chen and Pearl [50].
4) Source and target distributions may differ via covariate or target shift, but the DAG structure remains unchanged. In practice, such a structure can be learned from purely observational data using standard causal discovery methods [51], and further refined or validated when interventional measurements are available [52].

Domain Adaptation Task. Given

$$\mathcal{D}_{\text{s}} = \{\, X_{\text{s}}^{(i)} \}_{i=1}^{N_{\text{s}}}, \quad \mathcal{D}_{\text{t}}^{\text{obs}} = \{\, X_{\text{t},-t}^{(j)} \}_{j=1}^{N_{\text{t}}}, \quad \mathcal{G},$$

and shift assumptions, our goal is to impute $\{\, X_{t,\text{t}}^{(j)} \}_{j=1}^{N_{\text{t}}}$. Equivalently, we wish to compute

$$\widehat{X}_{t,\text{t}}^{(j)} \;=\; \mathbb{E}\big[X_t \mid X_{-t} = X_{\text{t},-t}^{(j)}\big], \quad j = 1, \ldots, N_{\text{t}},$$

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under the specified covariate/target shift conditions.

IV. Methodology and Theoretical Results

In this section, we first present the population-EM operator for the infinite-sample (no noise) setting, then describe a first-order (gradient) M–step, and finally give the sample-EM algorithm that jointly uses source and target data under domain shift. After outlining the algorithm, we develop the theoretical guarantees: population-level contraction, curvature decomposition, and sample-level error bounds.

A. Population-EM Operator under a Known Causal DAG

In the infinite-data regime (no sampling noise), the EM update can be written in closed form. Suppose the true joint distribution of X follows a linear Gaussian SEM with covariance $\Sigma^* \in \mathbb{R}^{p\times p}$. If $\Sigma^{(r)}$ is the current estimate, then we define the *negative* complete-data log-likelihood for a single observation X as

$$\ell_{\text{comp}}\big(X;\Sigma\big) \;=\; \tfrac{1}{2}\Big[\log\det(\Sigma) \;+\; \operatorname{trace}\big(\Sigma^{-1}XX^T\big)\Big].$$

Because $T = X_t$ is missing in the target domain, write $X_{-t} = (X_1,\dots,X_{t-1},X_{t+1},\dots,X_p)$. The population-level Q-function becomes

$$Q_{\text{pop}}\big(\Sigma \mid \Sigma^{(r)}\big) \;=\; -\,\mathbb{E}_{X_{-t}}\Big[\,\mathbb{E}\big[\ell_{\text{comp}}(X;\Sigma) \;\big|\; X_{-t},\,\Sigma^{(r)}\big]\Big].$$

Since ℓ_{comp} depends on $X\,X^T$, the inner expectation is

$$\mathbb{E}\big[X\,X^T \;\big|\; X_{-t}=x_{-t},\,\Sigma^{(r)}\big] = \begin{pmatrix} x_{-t}\,x_{-t}^T & x_{-t}\,\mu_t^{(r)} \\ \mu_t^{(r)}\,x_{-t}^T & V_t^{(r)} + \big(\mu_t^{(r)}\big)^2 \end{pmatrix}$$

where

$$\begin{aligned} \mu_t^{(r)} &= \Sigma_{t,-t}^{(r)}\big(\Sigma_{-t,-t}^{(r)}\big)^{-1}\,x_{-t}, \\ V_t^{(r)} &= \Sigma_{tt}^{(r)} \;-\; \Sigma_{t,-t}^{(r)}\big(\Sigma_{-t,-t}^{(r)}\big)^{-1}\,\Sigma_{-t,\,t}^{(r)}. \end{aligned} \tag{1}$$

Averaging over the marginal of X_{-t} yields

$$M^{(r)} \;=\; \mathbb{E}_{X_{-t}}\Big[\,\mathbb{E}\big[X\,X^T \mid X_{-t},\,\Sigma^{(r)}\big]\Big].$$

Hence

$$Q_{\text{pop}}\big(\Sigma \mid \Sigma^{(r)}\big) \;=\; -\,\tfrac{1}{2}\Big[\log\det(\Sigma) \;+\; \operatorname{trace}\big(\Sigma^{-1}M^{(r)}\big)\Big],$$

and its unique minimizer (equivalently, the maximizer of the true log-likelihood) is $\Sigma^{(r+1)} = M^{(r)}$. Thus the *population-EM operator* F is

$$F\big(\Sigma^{(r)}\big) \;=\; M^{(r)} \;=\; \mathbb{E}_{X_{-t}}\Big[\,\mathbb{E}\big[X\,X^T \mid X_{-t},\,\Sigma^{(r)}\big]\Big]. \tag{2}$$

In words, each iteration replaces the current covariance guess by the exact conditional second moment of X, averaged over the true distribution of X_{-t}. We will analyze the contraction properties of this mapping in Section IV-D.

B. First-Order (Partial) M–Step via Gradient-EM

In the finite-sample regime, the E–step produces a *completed* covariance estimate

$$\widehat{Q}^{(r)} \;=\; \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}\big[X^{(i)}X^{(i)\,T} \mid X_{-t}^{(i)},\,\Sigma^{(r)}\big],$$

where $n = N_{\text{t}}$ and X_{-t} denotes the observed coordinates. A standard M–step maximizes

$$\mathcal{L}\big(A,\Delta \mid \widehat{Q}^{(r)}\big) \;=\; -\frac{n}{2}\Big[\log\det\big(K(A,\Delta)\big) \;+\; \operatorname{trace}\big(K(A,\Delta)\,\widehat{Q}^{(r)}\big) \;+\; \sum_{i=1}^{p}\log\Delta_i\Big],$$

over the DAG parameters (A,Δ), with A the structural coefficient matrix and Δ the vector of noise variances where $K(A,\Delta)$ denotes the precision matrix induced by the DAG parameters:

$$K(A,\Delta) = A^T\operatorname{diag}(1/\Delta)\,A, \quad \Sigma(A,\Delta) = K(A,\Delta)^{-1}.$$

Rather than solve this $O(p^3)$ GLS subproblem exactly (i.e., by fitting (A,Δ) from $\widehat{Q}$ via generalized least squares), we take a single gradient-ascent step in (A,Δ), reducing each M–step to $O(p^2)$. Below we derive the required gradients.

DAG Parameterization. We parameterize the DAG by

$$A_{ii} = 1, \quad A_{ij} = -\theta_{ij} \text{ if } j \in \operatorname{pa}(i), \quad A_{ij} = 0 \text{ otherwise},$$

and $\Delta = (\Delta_1,\dots,\Delta_p)^T$ collects the noise variances. Define $\Lambda = \operatorname{diag}(1/\Delta)$, $K = A^T\Lambda A$, and $\Sigma = K^{-1}$.

(i) Gradient with respect to A. Write

$$\mathcal{L}(A,\Delta) = -\frac{n}{2}[\log\det(K) + \operatorname{trace}(K\,\widehat{Q}) + \sum_i \log\Delta_i].$$

Then

$$\frac{\partial}{\partial A}\log\det(K) \;=\; 2\,\Lambda\,A\,K^{-1}, \frac{\partial}{\partial A}\operatorname{trace}(K\,\widehat{Q}) \;=\; 2\,\Lambda\,A\,\widehat{Q}.$$

Hence (ignoring zero-entries outside $\operatorname{pa}(i)$) the ascent direction is

$$G_A \;=\; n\,\Lambda\,A\big[K^{-1} + \widehat{Q}\big].$$

(ii) Gradient with respect to Δ. Denoting the ith column of A by $A_{\cdot,i}$. Using $\partial_{\Delta_i}\log\det(K) = -\Delta_i^{-2}\,A_{\cdot,i}^T K^{-1}A_{\cdot,i}$ and $\partial_{\Delta_i}\operatorname{trace}(K\,\widehat{Q}) \;=\; -\Delta_i^{-2}\,A_{\cdot,i}^T\widehat{Q}\,A_{\cdot,i}$, plus $\partial_{\Delta_i}\sum_j\log\Delta_j \;=\; 1/\Delta_i$, one finds

$$\frac{\partial\mathcal{L}}{\partial\Delta_i} \;=\; -\frac{n}{2}\Big[-\Delta_i^{-2}\big(A_{\cdot,i}^T K^{-1}A_{\cdot,i}\big) - \Delta_i^{-2}\big(A_{\cdot,i}^T\widehat{Q}A_{\cdot,i}\big) + \frac{1}{\Delta_i}\Big].$$

Since $A_{\cdot,i}^T K^{-1}A_{\cdot,i} = \Sigma_{ii}$ and $\widehat{q}_{ii} = [\widehat{Q}]_{ii}$, the ascent direction is

$$G_{\Delta_i} \;=\; \frac{n}{2}\Big[\Delta_i^{-2}\big(\Sigma_{ii} + \widehat{q}_{ii}\big) \;-\; \Delta_i^{-1}\Big].$$

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First-Order Update Rules. At iteration r, given $(A^{(r)}, \Delta^{(r)})$, one can compute

$$\widehat{Q}^{(r)} \;=\; \frac{1}{N_{\mathrm{t}}} \sum_{i=1}^{N_{\mathrm{t}}} \mathbb{E}\big[X^{(i)} X^{(i)\,T} \mid X_{-t}^{(i)},\, \Sigma^{(r)}\big],$$

where $\Sigma^{(r)} = [\,K(A^{(r)}, \Delta^{(r)})\,]^{-1}$. Then evaluate

$$G_A = N_{\mathrm{t}}\, \Lambda^{(r)}\, A^{(r)} \Big[\Sigma^{(r)} + \widehat{Q}^{(r)}\Big],$$

$$G_{\Delta_i} = \frac{N_{\mathrm{t}}}{2} \Big[(\Delta_i^{(r)})^{-2} \big(\Sigma_{ii}^{(r)} + \widehat{q}_{ii}^{(r)}\big) \;-\; (\Delta_i^{(r)})^{-1}\Big].$$

For small step-sizes $\eta_A, \eta_\Delta > 0$, update

$$\widetilde{A} = A^{(r)} + \eta_A\, G_A, \quad \widetilde{\Delta}_i = \Delta_i^{(r)} + \eta_\Delta\, G_{\Delta_i}.$$

Then enforce sparsity and positivity: zero out $\widetilde{A}_{ij}$ if $j \notin \mathrm{pa}(i)$, and truncate $\widetilde{\Delta}_i \geq 10^{-6}$. Finally set

$$A^{(r+1)} = \widetilde{A}, \quad \Delta^{(r+1)} = \widetilde{\Delta},$$

$$\Sigma^{(r+1)} = \big[A^{(r+1)T} \mathrm{diag}(1/\Delta^{(r+1)})\, A^{(r+1)}\big]^{-1}.$$

This constitutes the *partial M–step*. Occasionally (e.g. every 50–100 iterations), one may run a full re-estimation of (A, Δ) to re-project onto the DAG-covariance manifold and maintain numerical stability.

C. Domain-Adaptive EM

In practice, we combine fully observed source data with partially observed target data via a two-stage EM procedure. A high-level description is as follows; detailed derivations and pseudo-code are deferred to the Supplementary Material.

1) Source-Domain Fit. Use the complete source samples to estimate all conditional parameters of the DAG that are invariant across domains. Concretely, letting

$$S_{\mathrm{s}} = \frac{1}{N_{\mathrm{s}}} \sum_{i=1}^{N_{\mathrm{s}}} X_{\mathrm{s}}^{(i)}\, X_{\mathrm{s}}^{(i)\,T},$$

$$(A^{(\mathrm{s})},\, \Delta^{(\mathrm{s})}) = \mathrm{FitDAG}(A,\, S_{\mathrm{s}},\, N_{\mathrm{s}})\,,$$

where FitDAG denotes the generalized least squares procedure that estimates the structural coefficients A and noise variances Δ from the sample covariance S_{s} under the given DAG structure. We identify which rows/columns of $(A^{(\mathrm{s})}, \Delta^{(\mathrm{s})})$ remain fixed under covariate or target shift:

- Under *covariate shift*, all edges and noise variances are invariant, so $(A^{(\mathrm{s})}, \Delta^{(\mathrm{s})})$ is frozen entirely.
- Under *target shift*, only the conditional of T may change, so $(\,A_{-t,-t}^{(\mathrm{s})},\, \Delta_{-t}^{(\mathrm{s})}\,)$ remain fixed.

2) Target-Domain EM. We perform EM on the target observed-data likelihood, holding the source-learned parameters fixed except for those affected by the shift. At each iteration r:

- **E–step (target).** Compute the "completed" covariance

$$Q_{\mathrm{tgt}}^{(r)} \;=\; \frac{1}{N_{\mathrm{t}}} \sum_{j=1}^{N_{\mathrm{t}}} \mathbb{E}\big[X_{\mathrm{t}}^{(j)} X_{\mathrm{t}}^{(j)\,T} \;\big|\; X_{-t,\mathrm{t}}^{(j)},\, \Sigma^{(r)}\big],$$

  using standard Gaussian–conditioning formulas.
- **M–step (target).** Maximize the hybrid "source + target" Q-function over only the parameters affected by shift, keeping invariant rows/columns fixed to $(A^{(\mathrm{s})}, \Delta^{(\mathrm{s})})$. In particular:
  - Under *covariate shift*, update $\big(A_{t,-t},\, \Delta_t\big)$ via a simple regression of X_t on $X_{\mathrm{pa}(t)}$ using the block of $Q_{\mathrm{tgt}}^{(r)}$.
  - Under *target shift*, update only Δ_t (the marginal variance of T) via its completed-data second moment.

  Reconstruct $\Sigma^{(r+1)}$ from the updated $(A^{(r+1)},\, \Delta^{(r+1)})$.
- **Convergence check.** Stop when $\|\Sigma^{(r+1)} - \Sigma^{(r)}\|_F$ falls below a threshold ε.

Remarks.

- Each target M–step now requires only $O(p^2)$ computations (a small regression or variance update), rather than a full $O(p^3)$ DAG-fit.
- Under pure covariate shift, all parameters except $(A_{t,-t},\, \Delta_t)$ remain locked to the source estimates.
- Under pure target shift, the M–step reduces to updating Δ_t alone.

D. Population-Level Contraction in a Neighborhood of Σ^*

Let Σ^* denote the true target covariance, i.e. the unique fixed point of the population EM operator F in (2). We will show that there exists $r > 0$ and a constant $0 \leq \kappa < 1$ such that

$$\big\|F(\Sigma) - \Sigma^*\big\|_F \;\leq\; \kappa\, \|\Sigma - \Sigma^*\|_F, \quad \forall\, \Sigma \in \mathcal{B}_F(\Sigma^*; r),$$

where $\mathcal{B}_F(\Sigma^*; r) = \{\Sigma \succ 0 : \|\Sigma - \Sigma^*\|_F \leq r\}$.

(a) Preliminaries. Denote by $\mathcal{M}_+^p$ the set of $p \times p$ positive-definite matrices, equipped with the Frobenius norm $\|\cdot\|_F$. The EM operator

$$F : \mathcal{M}_+^p \to \mathcal{M}_+^p, \quad F(\Sigma) = \mathbb{E}_{X_{-t}}\Big[\mathbb{E}(X\, X^T \mid X_{-t}, \Sigma)\Big].$$

We make two assumptions:

Assumption 1 (Strong-Concavity of Complete-Data Likelihood): There exists $\lambda > 0$ such that, in a neighborhood of Σ^*, the negative complete-data log-likelihood $\ell_{\mathrm{comp}}(X; \Sigma) = \frac{1}{2}[\log\det(\Sigma) + X^T \Sigma^{-1} X]$ is λ-strongly convex in Σ^{-1}. Equivalently,

$$\nabla^2_{\Sigma^{-1}} \mathbb{E}[\ell_{\mathrm{complete}}(X; \Sigma)]\big|_{\Sigma=\Sigma^*} \;\succeq\; \lambda\, I_{p(p+1)/2}.$$

Assumption 2 (Lipschitz Missing-Data Term): In the same neighborhood, the missing-data Hessian

$$\Delta(\Sigma) \;=\; \mathbb{E}_{X_{-t}}\Big[\nabla^2_{\Sigma^{-1}} \log p(X_{-t}; \Sigma)\Big]$$

is Lipschitz in Frobenius norm with constant $\gamma \geq 0$. That is, for all $\Sigma_1, \Sigma_2 \in \mathcal{B}_F(\Sigma^*; r)$,

$$\big\|\Delta(\Sigma_1) - \Delta(\Sigma_2)\big\|_F \;\leq\; \gamma\, \big\|\Sigma_1 - \Sigma_2\big\|_F.$$

Under these assumptions (so that $\gamma < \lambda$), standard analysis (e.g. [34]) implies that F is locally Lipschitz with constant $\kappa = \gamma/\lambda < 1$, yielding contraction.

(b) Contraction Theorem.

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Theorem 1 (Population-EM Contraction): Suppose Assumptions 1 & 2 hold with $\gamma < \lambda$. Then there exists $r > 0$ s.t.

$$\left\|F(\Sigma) - \Sigma^*\right\|_F \;\le\; \kappa \left\|\Sigma - \Sigma^*\right\|_F, \quad \forall\, \Sigma \in \mathcal{B}_F(\Sigma^*; r),$$

where $\kappa = \gamma/\lambda < 1$. Consequently, Σ^* is the unique fixed point in $\mathcal{B}_F(\Sigma^*; r)$, and iterates $\Sigma^{(r+1)} = F(\Sigma^{(r)})$ converge geometrically whenever $\|\Sigma^{(0)} - \Sigma^*\|_F \le r$.

Proof Sketch: By Taylor expanding the population Q-function around Σ^*, one can show that $\|F(\Sigma) - F(\Sigma^*)\|_F \le (\gamma/\lambda)\,\|\Sigma - \Sigma^*\|_F$ for Σ sufficiently close to Σ^*. Since $F(\Sigma^*) = \Sigma^*$, the contraction follows. See [34] for full details adapted to our DAG setting. ∎

(c) Choice of Radius. A valid choice is

$$r \;=\; \frac{\lambda - \gamma}{L},$$

where L bounds the third derivative of ℓ_{comp} in a neighborhood of Σ^*. In practice, one ensures $\|\Sigma^{(0)} - \Sigma^*\|_F \le r$ via a pilot-EM or covariate-shift initialization (Section IV-C).

(d) Implications under Domain Shift. Under pure covariate shift, λ (complete-data curvature) remains unchanged, while γ depends only on $\Sigma^*_{-t,-t}$. Hence if $\lambda > \gamma$ in the source, it holds in the target regardless of how covariates change. Under partial target shift, γ may increase modestly, but provided Σ^* remains well-conditioned (i.e. $\lambda_{\min}(\Sigma^*)$ stays bounded away from zero), one still has $\gamma < \lambda$. Therefore, geometric-rate convergence holds uniformly across a range of covariate/target shift scenarios.

E. EM Curvature Decomposition

To verify $\gamma < \lambda$ explicitly, we decompose the Hessian of the observed-data negative log-likelihood into a complete-data term minus a missing-data correction (cf. [36]). Let $X = (X_1, \ldots, X_p)^T$ and write the negative complete-data log-likelihood as $\ell_{\text{comp}}(X; \Sigma)$ so that $\nabla^2_{\Sigma^{-1}} \ell_{\text{comp}}(X; \Sigma) = \frac{1}{2}\,(\Sigma \otimes \Sigma)$, where the symbol $\otimes$ is the Kronecker product. Hence if $X \sim N(0, \Sigma)$, then

$$\mathbb{E}\big[\nabla^2_{\Sigma^{-1}} \ell_{\text{comp}}(X; \Sigma)\big] \;=\; \tfrac{1}{2}\,(\Sigma \otimes \Sigma).$$

When T is missing, the observed-data negative log-likelihood (i.e. the negative log of the marginal density of X_{-t}) is

$$\ell_{\text{obs}}(X_{-t}; \Sigma) \;=\; -\log \int \exp\big(-\ell_{\text{comp}}(X; \Sigma)\big)\, dX_t.$$

Its Hessian with respect to Σ^{-1}, evaluated at $\Sigma = \Sigma^*$, decomposes as

$$\nabla^2_{\Sigma^{-1}} \ell_{\text{obs}}\big(X_{-t}; \Sigma^*\big) \;=\; H_{\text{comp}} \;-\; H_{\text{miss}}, \tag{3}$$

where

$$H_{\text{comp}} \;=\; \mathbb{E}_X\big[\nabla^2_{\Sigma^{-1}} \ell_{\text{comp}}(X; \Sigma^*)\big] \;=\; \tfrac{1}{2}\,(\Sigma^* \otimes \Sigma^*),$$

and

$$H_{\text{miss}} \;=\; \mathbb{E}_{X_{-t}}\Big[\nabla^2_{\Sigma^{-1}} \log p(X_{-t}; \Sigma)\Big]_{\Sigma = \Sigma^*}.$$

Below, we derive H_{miss} in closed form.

(a) Form of the Missing-Data Hessian. Partition $\Sigma = \begin{pmatrix} \Sigma_{-t,-t} & \Sigma_{-t,t} \\ \Sigma_{t,-t} & \Sigma_{t,t} \end{pmatrix}$. Since

$$p(X_{-t}; \Sigma) = \frac{\exp\Big(-\frac{1}{2} X_{-t}^T \Sigma^{-1}_{-t,-t} X_{-t}\Big)}{(2\pi)^{(p-1)/2} \sqrt{\det(\Sigma_{-t,-t})}},$$

its negative log-marginal is

$$\ell_{\text{miss}}(X_{-t}; \Sigma) = \tfrac{1}{2}[\log\det(\Sigma_{-t,-t}) + X_{-t}^T \Sigma^{-1}_{-t,-t} X_{-t}]$$

Differentiating twice with respect to $\Psi = \Sigma^{-1}$ yields

$$\nabla^2_{\Psi} \log p\big(X_{-t}; \Sigma\big) = \frac{1}{2}\Big[\, E_{-t,-t}\, \Sigma_{-t,-t}\, E_{-t,-t}^T \;-\; E_{-t,-t}\, X_{-t} X_{-t}^T\, E_{-t,-t}^T\Big],$$

where $E_{-t,-t}$ is the $(p \times (p-1))$ selection matrix extracting rows and columns corresponding to $\{1, \ldots, p\} \setminus \{t\}$. Taking expectation over $X_{-t} \sim N(0, \Sigma^*_{-t,-t})$ gives

$$H_{\text{miss}} = \mathbb{E}_{X_{-t}}\Big[\nabla^2_{\Psi} \log p(X_{-t}; \Sigma^*)\Big] = \tfrac{1}{2}\, E_{-t,-t}\, \Sigma^*_{-t,-t}\, E_{-t,-t}^T,$$

a positive semidefinite matrix of rank at most $p - 1$.

(b) Contraction Constant. Substituting into (3),

$$\nabla^2_{\Sigma^{-1}} \ell_{\text{obs}}\big(X_{-t}; \Sigma^*\big) = \tfrac{1}{2}\Big[\Sigma^* \otimes \Sigma^* - E_{-t,-t}\, \Sigma^*_{-t,-t}\, E_{-t,-t}^T\Big].$$

Restricted to the DAG-constraint subspace in Σ^{-1}, the smallest eigenvalue is at least

$$\frac{1}{2}\Big[\lambda_{\min}(\Sigma^*)^2 - \lambda_{\max}(\Sigma^*_{-t,-t})\Big].$$

Define

$$\lambda = \tfrac{1}{2}\left[\lambda_{\min}(\Sigma^*)\right]^2, \quad \gamma = \tfrac{1}{2}\,\lambda_{\max}\big(\Sigma^*_{-t,-t}\big),$$
$$\implies \quad \frac{\gamma}{\lambda} = \frac{\lambda_{\max}(\Sigma^*_{-t,-t})}{\left[\lambda_{\min}(\Sigma^*)\right]^2}. \tag{4}$$

Whenever $\gamma/\lambda < 1$, the observed-data Hessian is strictly positive definite in Σ^{-1}, implying $\kappa = \gamma/\lambda < 1$ and local contraction of F.

(c) Effect of Covariate and Target Shift.

- Under *pure covariate shift*, $p(X_t \mid X_{-t})$ is invariant, so the T-row/column of Σ^* remains the same; only $\Sigma^*_{-t,-t}$ changes. Hence $\lambda_{\max}(\Sigma^*_{-t,-t})$ may vary, but $\lambda_{\min}(\Sigma^*)$ stays bounded by its original source value. As long as
$$\lambda_{\max}\big(\Sigma^*_{-t,-t}\big) \;<\; \left[\lambda_{\min}(\Sigma^*)\right]^2,$$
contraction holds.
- Under *pure target shift*, $\Sigma^*_{-t,-t}$ is unchanged, so $\lambda_{\max}(\Sigma^*_{-t,-t})$ stays fixed. However, $\lambda_{\min}(\Sigma^*)$ may decrease if $\Sigma^*_{t,t}$ changes drastically. To maintain $\gamma < \lambda$, one requires
$$\lambda_{\min}(\Sigma^*) \;\ge\; \sqrt{\lambda_{\max}\big(\Sigma^*_{-t,-t}\big)}.$$
- Under *combined covariate & target shift*, both blocks may change. The general requirement remains
$$\lambda_{\max}\big(\Sigma^*_{-t,-t}\big) \;<\; \left[\lambda_{\min}(\Sigma^*)\right]^2,$$

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where Σ^* is the final target covariance. Verifying this spectral-gap condition ensures geometric convergence under any allowable shift.

Thus, the curvature decomposition provides a concrete criterion—namely $\lambda_{\max}(\Sigma^*_{-t,-t}) < [\lambda_{\min}(\Sigma^*)]^2$—to check for geometric contraction under covariate and/or target shift.

F. High-Probability Sample-Level Concentration and Final Error Bound

Having established population-level contraction ($\kappa < 1$), we turn to finite samples. Let $\widehat{F}$ denote the sample-EM operator that replaces population expectations by their empirical counterparts. Under standard concentration bounds for sub-Gaussian vectors and Weyl's inequality for eigenvalues, one shows that, with high probability,

$$\left\|\widehat{F}(\Sigma) - F(\Sigma)\right\|_F \;\le\; \delta(N_{\mathrm{s}}, N_{\mathrm{t}}), \quad \forall\, \Sigma \in \mathcal{B}_F(\Sigma^*; r),$$

where $\delta(\cdot) \to 0$ as $N_{\mathrm{s}}, N_{\mathrm{t}} \to \infty$. Combining this with population contraction $\|F(\Sigma) - \Sigma^*\|_F \le \kappa \|\Sigma - \Sigma^*\|_F$, one shows that the sample iterates $\Sigma^{(r)}$ satisfy

$$\|\Sigma^{(r)} - \Sigma^*\|_F \;\le\; \kappa^{\,r} \|\Sigma^{(0)} - \Sigma^*\|_F \;+\; \frac{\delta}{1-\kappa}.$$

Hence, for a suitable initialization within the contraction ball, the finite-sample EM converges rapidly to an $O(\delta/(1-\kappa))$-neighborhood of Σ^* with high probability. Detailed proofs follow the template of [34] adapted to the DAG setting under domain shift.

G. Other EM Variants with Geometric-Rate Guarantees

Beyond the proposed 1st-order EM, it is natural to ask whether related EM-family algorithms also enjoy local geometric convergence in our setting. Below we describe three such variants. For the Exact EM and also for the ECME [42], formal contraction proofs can be established under the same curvature and spectral-gap conditions used in our analysis. For PX-EM [53], a suitable parameter expansion yields the same type of guarantee, though a complete proof requires additional technical development. Full details and preliminary experimental evidence are provided in the supplementary materials.

a) Exact EM with block restriction: If initialized within the contraction basin (e.g., using the source estimate) and restricted to updating only the shifted block $(A_{t,-t}, \Delta_t)$ while freezing invariants, the standard EM operator admits the same local contractivity as our 1st-order EM analysis. This corresponds to an ECM-style (block-coordinate) update [54]: closed-form regression for $A_{t,-t}$ and a variance update for Δ_t using the completed covariance. Compared with 1st-order EM, each iteration is more expensive (involving generalized least squares and DAG fitting), but fewer steps may suffice once inside the basin.

b) ECME (Expectation/Conditional Maximization Either): Replacing the Δ_t update by a direct maximization of the observed-data likelihood, while keeping the E-step unchanged and freezing invariants, yields an ECME operator. Under the same local strong concavity and spectral-gap assumptions, one can formally prove that ECME is contractive in a neighborhood of Σ^*, with a rate constant no worse than that of EM. In practice, ECME often enlarges the basin of attraction and accelerates convergence.

c) PX-EM (Parameter-Expanded EM): Parameter expansion followed by smooth re-projection can reduce the fraction of missing information, thereby improving the contraction constant κ. With an appropriate scale-expansion (e.g., jointly scaling $A_{t,-t}$ and Δ_t), the PX-EM operator inherits local contractivity from the expanded space and achieves geometric convergence with a rate constant no worse—and often strictly better—than plain EM. As with the other variants, the guarantee is local and depends on the same curvature and spectral-gap conditions. A complete proof in our setting requires verifying smooth invertibility of the expansion–projection mapping and bounding the expanded Jacobian; these technical details are outlined in the supplementary materials.

V. Experimental Results

We evaluate the performance of our 1st-order EM algorithm for fitting Gaussian DAG models with a single latent variable, comparing it against the classical Kiiveri EM procedure and a standard Bayesian network baseline. Our study includes two parts: (i) controlled experiments on simulated datasets, where we assess robustness and accuracy across domains under varying DAG sizes and shift scenarios, and (ii) a real-data case study on single-cell signaling measurements, which provides a stringent test of transfer in a biological system. All experiments were executed on a Windows workstation equipped with a 12th Gen Intel(R) Core(TM) i9-12900H 2.50 GHz processor. The source code and experimental scripts for this work are publicly available at https://github.com/majavid/ICDM2025.

A. Simulated Experiments

Referring back to Section I and the motivating example, we consider a seven-node SEM in which two context variables C_1, C_2 drive intermediate nodes Z and X, which together with C_1 determine the treatment node T, and T in turn influences outcomes P and Y. Under covariate shift (a large change in the marginal of C_2) or target shift (a large change in the marginal of C_1), we "hide" T in the target domain and impute it from the observed parents $\{C_1, X, Z\}$. This setup pits three methods against each other: (i) the BN baseline that directly applies source-learned regression coefficients, (ii) the Kiiveri EM that iteratively estimates the complete-data covariance with one latent, and (iii) our single–update 1st-order EM.

Our results (Table 2) show that the BN baseline remains reasonably accurate under covariate shift but degrades under target shift, while the Kiiveri EM routinely fails in both scenarios, producing extremely large errors. In contrast, the 1st-order EM achieves consistently low MAE and RMSE (on

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the order of 10^{-1}) and near-perfect R^2 (≈ 0.999) in both covariate and target shifts. These findings demonstrate that a single–step EM correction is sufficient to adapt to major distributional changes in the parent variables, combining the simplicity of conditional regression with the flexibility of latent-variable modeling.

TABLE 2. Average error metrics under covariate and target shift (10 repeats).

Shift scenario	Method	MAE	RMSE	R^2
Covariate shift	Baseline (Fit-on-Source)	0.7935	0.9945	0.9981
	Kiiveri EM	45.1882	45.1973	–2.9821
	1st-order EM	**0.3299**	**0.4145**	**0.9997**
Target shift	Baseline (Fit-on-Source)	6.0107	6.3333	0.9473
	Kiiveri EM	70.8294	72.1688	–5.8331
	1st-order EM	**0.9312**	**1.0577**	**0.9985**

TABLE 3. Imputation performance on the MAGIC-IRRI DAG under multiple large-shift interventions.

Method	MAE	RMSE	R^2
Baseline (Fit-on-Source)	9.3827	11.1872	–0.0957
Kiiveri EM	8.8479	11.0771	–0.0743
1st-order EM	**5.5834**	**7.0277**	**0.5676**

In another experiment on the 64-node MAGIC-IRRI Gaussian Bayesian network (Scutari ICQG 2016)[2], we induced large "marginal-shift" interventions on three continuous variables:

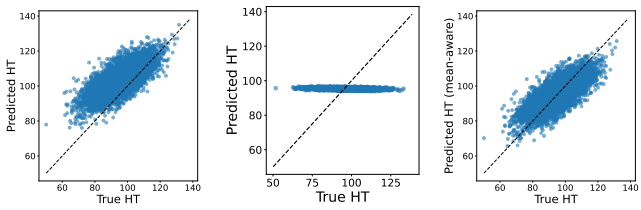
- **G4156**: from $N(0.7636,\ 0.9721^2)$ to $N(1.5,\ 2.0^2)$.
- **G4573**: from $N(0.1196,\ 0.4744^2)$ to $N(1.0,\ 1.0^2)$.
- **G1533**: from $N(0.8004,\ 0.9803^2)$ to $N(0,\ 3.0^2)$.

These shifts emulate strong experimental perturbations in a multi-trait genetic model. We then hid the target variable `HT` in the shifted domain and attempted imputation using: (i) a source-only linear fit ("Baseline"), (ii) the classical Kiiveri EM for one latent, and (iii) our 1st-order EM.

As shown in Table 3, the baseline fit-on-source regression already struggles under large distribution shifts (MAE ≈ 9.38, RMSE ≈ 11.19, $R^2 \approx -0.10$), highlighting its inability to adapt to shifted marginals. Kiiveri's full EM offers only a marginal improvement (MAE ≈ 8.85, RMSE ≈ 11.08, $R^2 \approx -0.07$), suggesting that overfitting the latent structure provides little benefit under extreme interventions. In sharp contrast, our 1st-order EM drastically reduces errors (MAE ≈ 5.58, RMSE ≈ 7.03) and achieves a positive $R^2 \approx 0.57$, demonstrating that even a single covariance correction is sufficient to recover substantial predictive power in high-dimensional, heavily perturbed Gaussian DAGs.

Figure 3 illustrates the performance of the three imputation methods under a severe covariate shift. In panel (a), the Fit-on-Source baseline shows a wide scatter and systematic bias away from the $y = x$ reference line, consistent with its large errors and negative R^2. Panel (b) demonstrates that

[2]The network structure and data are available at https://www.bnlearn.com/bnrepository/.



(a) Baseline (BN) (b) Kiiveri EM (c) 1st-order EM

Fig. 3. True versus predicted HT under covariate shift for the three methods: (a) Fit-on-Source baseline, (b) classical Kiiveri EM, (c) our 1st-order EM.

the classical Kiiveri EM likewise fails to adapt, collapsing predictions toward a narrow range and thereby increasing distortion under intervention. By contrast, panel (c) shows that our 1st-order EM concentrates predictions much more tightly around the diagonal, markedly reducing both bias and spread. Quantitatively, the 1st-order EM cuts MAE and RMSE by more than one third relative to the others and attains a positive R^2 of about 0.57, confirming its robustness to heavy-shift perturbations in high-dimensional Gaussian DAGs.

B. Real-Data Experiment on Single-Cell Signaling Networks

To evaluate our method in a realistic setting, we applied it to the landmark single-cell flow cytometry dataset of Sachs et al. [55], which measures phosphorylated signaling proteins in human primary $CD4^+$ T cells under targeted perturbations. This dataset is well-suited for causal transfer learning because it combines observational variability with experimental interventions, thereby inducing domain shifts across conditions. For our evaluation, we designated the anti-CD3/CD28 stimulation condition (853 cells) as the source domain and the PMA stimulation condition (913 cells) as the target domain. This setup provides a stringent test of transfer, since PMA acts as a direct PKC activator that alters downstream signaling dynamics in ways not present under CD3/CD28 stimulation. Using this source–target split, we focused on predicting the activity of ten signaling targets (Raf, Mek, Plcg, PIP_2, PIP_3, Erk, Akt, PKA, P38, and Jnk). Table 4 reports predictive performance in terms of MAE, RMSE, and coefficient of determination (R^2).

The results indicate substantial improvements for several proteins when transferring from source to target. For Raf and Plcg, predictive accuracy improved markedly, with R^2 values increasing from near zero to 0.59 and 0.19, respectively. The most striking gain was observed for Erk, where the R^2 rose to 0.92, highlighting the ability of our framework to leverage interventional variability for robust inference. Smaller but consistent improvements were also observed for PIP_3 (from 0.14 to 0.34). For Akt, performance was strong across all approaches, with $R^2 \approx 0.92$, suggesting that this protein's dynamics are well captured under both conditions.

At the same time, performance deteriorated for certain targets such as Mek, PKA, and Jnk, where negative R^2 values were obtained. These failures likely reflect mismatches

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TABLE 4. Imputation performance on the Sachs et al. data under multiple interventions.

Target Variable	Method	MAE	RMSE	R2
Raf	Baseline (BN)	0.6908	1.0015	-0.0041
	1st-order EM	**0.4132**	**0.6393**	**0.5908**
	Kiiveri-EM	0.5132	0.8324	0.3064
Mek	Baseline (BN)	**0.3933**	**0.6383**	**0.5922**
	1st-order EM	0.7140	1.0701	-0.1464
	Kiiveri-EM	0.7143	1.0707	-0.1476
Plcg	Baseline (BN)	0.6529	0.9995	-0.0000
	1st-order EM	0.5858	0.9008	0.1876
	Kiiveri-EM	**0.5851**	**0.8998**	**0.1894**
PIP2	Baseline (BN)	0.6156	0.8254	0.3180
	1st-order EM	0.6156	0.8254	0.3180
	Kiiveri-EM	0.6165	0.8263	0.3165
PIP3	Baseline (BN)	0.5240	0.9280	0.1378
	1st-order EM	0.3809	0.8106	0.3422
	Kiiveri-EM	**0.3731**	**0.8049**	**0.3515**
Erk	Baseline (BN)	0.5884	0.8379	0.2971
	1st-order EM	**0.1817**	**0.2855**	**0.9184**
	Kiiveri-EM	4.7827	6.8503	-45.9786
Akt	Baseline (BN)	0.1744	0.2756	0.9240
	1st-order EM	0.1744	0.2756	0.9240
	Kiiveri-EM	**0.1728**	**0.2742**	**0.9247**
PKA	Baseline (BN)	**0.6810**	**0.9992**	**0.0006**
	1st-order EM	0.7594	1.0982	-0.2073
	Kiiveri-EM	0.8031	1.1795	-0.3927
P38	Baseline (BN)	0.2897	0.4543	0.7934
	1st-order EM	0.2897	0.4543	0.7934
	Kiiveri-EM	0.2897	0.4543	0.7934
Jnk	Baseline (BN)	0.6621	1.1185	-0.2525
	1st-order EM	0.6621	1.1185	-0.2525
	Kiiveri-EM	0.6620	1.1185	-0.2525

between the Gaussian DAG assumptions of our framework and the complexities of biological signaling. Non-Gaussian noise, hidden confounders, and feedback loops are all known features of cellular networks that can undermine theoretical guarantees. Moreover, PMA directly perturbs PKC activity, which can propagate nonlinear shifts that destabilize transfer. Such cases underscore that, while the method is effective in structured domains, it remains sensitive to structural misspecification in heterogeneous biological data.

Overall, this experiment provides both validation and caution. Strong results for Raf, Erk, and Plcg confirm that the framework can successfully exploit cross-condition data for improved prediction, while weaker performance for Mek and PKA highlights the need for more robust extensions that can handle feedback, hidden variables, and non-Gaussianity. Future work in this direction may further enhance the reliability of causal transfer methods in complex biological systems.

VI. Conclusion

In this paper, we introduced a unified EM-based framework for imputing a designated target variable in Gaussian causal models under both covariate and target domain shifts. Building on a known DAG structure, we compared a simple fit-on-source regression, the classical Kiiveri EM algorithm, and our 1st-order EM—which performs just a single complete-data covariance correction. Across experiments on a synthetic seven-node SEM, the 64-node MAGIC-IRRI network, and the real single-cell signaling dataset of Sachs et al., we showed that the 1st-order EM achieves near-oracle accuracy under severe shifts in the small model, substantially outperforms alternatives in the large high-dimensional setting, and delivers strong predictive gains for key signaling proteins in biological data—all while scaling linearly with the number of edges.

Looking forward, there are several promising directions. First, although our current method assumes exactly one latent (missing) variable, the underlying theory can be extended to multiple hidden nodes in a principled way, by generalizing the 1st-order covariance correction to block-latent updates. Second, we have assumed causal sufficiency and a fully observed DAG structure; in many real applications, one must contend with latent confounders and selection bias, which are properly encoded by acyclic directed mixed graphs (ADMGs) or ancestral graphs. Extending our EM framework to these richer graphical models will require new conditional-moment calculations and careful study of convergence guarantees. Finally, a rigorous analysis of the EM algorithm's convergence rate and statistical consistency under model misspecification remains an important theoretical challenge. We leave these extensions for future work.

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